

Date / /
 Any tangent to the auxiliary circle is the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

hence proved.

Q. S.T locus of poles of normal chords of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is the curve $y^2 a^6 - x^2 b^6 = (a^2 + b^2)^2 x^2 y^2$

Let the normal chord be PQ at $(a \sec \theta, b \tan \theta)$

Eqn of PQ is

$$\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2 \quad \text{--- (1)}$$

Let (x_1, y_1) be the pole of eqn (1). Then PQ is the polar of (x_1, y_1) and also eqn of polar is

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \quad \text{--- (2)}$$

eqn (1) and (2) represent same line.

$$\frac{a/\sec \theta}{x_1/a^2} = \frac{b/\tan \theta}{-y_1/b^2} = \frac{a^2 + b^2}{1}$$

$$\frac{a^3}{x_1 \sec \theta} = a^2 + b^2$$

$$\frac{b^3}{-y_1 \tan \theta} = a^2 + b^2$$

$$\Rightarrow \sec \theta = \frac{a^3}{(a^2 + b^2) x_1}$$

$$\Rightarrow \tan \theta = \frac{-b^3}{y_1 (a^2 + b^2)}$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\left(\frac{a^3}{(a^2+b^2)x_1} \right)^2 - \left(\frac{b^3}{y_1(a^2+b^2)} \right)^2 = 1$$

$$\frac{a^6}{x_1^2} - \frac{b^6}{y_1^2} = (a^2+b^2)^2$$

$$\underline{y_1^2 a^6 - b^6 x_1^2 = x_1^2 y_1^2 (a^2 + b^2)^2}$$

∴ locus of poles of normal chords is the curve
 $\underline{y^2 a^6 - x^2 b^6 = (a^2 + b^2)^2 x^2 y^2}$